

Overview of materials for Polypropylene, Molded

Categories: [Polymer](#); [Thermoplastic](#); [Polypropylene \(PP\)](#); [Polypropylene, Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polypropylene, Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

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Physical Properties	Metric	English	Comments
Density	0.880 - 2.40 g/cc	0.0318 - 0.0867 lb/in ³	Average value: 0.931 g/cc Grade Count:1199
Water Absorption	0.000 - 0.800 %	0.000 - 0.800 %	Average value: 0.0663 % Grade Count:114
Moisture Absorption at Equilibrium	0.100 %	0.100 %	Average value: 0.100 % Grade Count:61
Water Absorption at Saturation	0.100 %	0.100 %	Average value: 0.100 % Grade Count:3
Particle Size	300 - 1000 µm	300 - 1000 µm	Average value: 675 µm Grade Count:4
Maximum Moisture Content 	0.250 - 0.250 @Temperature 120 - 120 °C	0.250 - 0.250 @Temperature 248 - 248 °F	Average value: 0.250 Grade Count:1
	0.250 - 0.250 @Time 299 - 299 sec	0.250 - 0.250 @Time 0.0830 - 0.0830 hour	Average value: 0.250 Grade Count:1
Linear Mold Shrinkage 	0.000500 - 0.0250 cm/cm	0.000500 - 0.0250 in/in	Average value: 0.0134 cm/cm Grade Count:240
	0.0150 - 0.0200 cm/cm @Temperature 80.0 - 80.0 °C	0.0150 - 0.0200 in/in @Temperature 176 - 176 °F	Average value: 0.0175 cm/cm Grade Count:3
Linear Mold Shrinkage, Transverse 	0.00600 - 0.0200 cm/cm	0.00600 - 0.0200 in/in	Average value: 0.0128 cm/cm Grade Count:33
	0.0150 - 0.0200 cm/cm @Temperature 80.0 - 80.0 °C	0.0150 - 0.0200 in/in @Temperature 176 - 176 °F	Average value: 0.0175 cm/cm Grade Count:3
Melt Flow	0.200 - 1800 g/10 min	0.200 - 1800 g/10 min	Average value: 28.9 g/10 min Grade Count:1322
Base Resin Melt Index	0.500 - 50.0 g/10 min @Temperature 230 - 230 °C	0.500 - 50.0 g/10 min @Temperature 446 - 446 °F	Average value: 14.9 g/10 min Grade Count:19
Ash 	0.000 - 26.0 %	0.000 - 26.0 %	Average value: 1.83 % Grade Count:17
	57.0 - 63.0 % @Temperature 625 - 625 °C	57.0 - 63.0 % @Temperature 1160 - 1160 °F	Average value: 14.2 % Grade Count:6
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	20.0 - 118	20.0 - 118	Average value: 94.6 Grade Count:594
Hardness, Shore D	30.0 - 83.0	30.0 - 83.0	Average value: 67.5 Grade Count:111
Ball Indentation Hardness	45.0 - 106 MPa	6530 - 15400 psi	Average value: 72.5 MPa Grade Count:37

Tensile Strength, Ultimate	9.00 - 80.0 MPa	1310 - 11600 psi	Average value: 29.5 MPa Grade Count:207
Tensile Strength, Yield	4.00 - 369 MPa	580 - 53500 psi	Average value: 31.5 MPa Grade Count:1184
Film Elongation at Break, MD	93.0 - 530 %	93.0 - 530 %	Average value: 258 % Grade Count:3
Elongation at Break	2.40 - 900 %	2.40 - 900 %	Average value: 159 % Grade Count:421
 @Temperature -17.7 - -17.7 °C	8.00 - 14.0 %	8.00 - 14.0 % @Temperature 0.126 - 0.126 °F	Average value: 9.70 % Grade Count:10
Elongation at Yield	1.80 - 160 %	1.80 - 160 %	Average value: 10.5 % Grade Count:869
Modulus of Elasticity	0.00800 - 8.25 GPa	1.16 - 1200 ksi	Average value: 1.69 GPa Grade Count:334
Tenacity	0.203 - 0.397 N/tex	2.30 - 4.50 g/denier	Average value: 0.302 N/tex Grade Count:12
Flexural Yield Strength	20.0 - 180 MPa	2900 - 26100 psi	Average value: 44.2 MPa Grade Count:69
Flexural Modulus	0.0260 - 10.0 GPa	3.77 - 1450 ksi	Average value: 1.43 GPa Grade Count:1090
 @Temperature 90.0 - 90.0 °C	0.410 - 0.410 GPa	59.5 - 59.5 ksi @Temperature 194 - 194 °F	Average value: 0.410 GPa Grade Count:1
Shear Modulus	0.680 - 0.920 GPa	98.6 - 133 ksi	Average value: 0.757 GPa Grade Count:10
Secant Modulus	1.14 - 1.97 GPa	165 - 286 ksi	Average value: 1.47 GPa Grade Count:19
Izod Impact	8.00 - 10.0 J	5.90 - 7.38 ft-lb	Average value: 9.00 J Grade Count:3
Izod Impact, Notched	0.0170 J/cm - NB	0.0318 ft-lb/in - NB	Average value: 0.732 J/cm Grade Count:894
 @Temperature -40.0 - 0.000 °C	0.0400 - 99.99 J/cm	0.0749 - 187.3 ft-lb/in @Temperature -40.0 - 32.0 °F	Average value: 0.522 J/cm Grade Count:96
 @Temperature -20.0 - -20.0 °C	1.00 - 1.35 J/cm	1.87 - 2.53 ft-lb/in @Temperature -4.00 - -4.00 °F	Average value: 0.522 J/cm Grade Count:2
	1.00 - 1.35 J/cm @Diameter 3.17 - 3.17 mm	1.87 - 2.53 ft-lb/in @Diameter 0.125 - 0.125 in	Average value: 0.522 J/cm Grade Count:2
Izod Impact, Unnotched	0.196 J/cm - NB	0.367 ft-lb/in - NB	Average value: 7.26 J/cm Grade Count:67
 @Temperature -18.0 - -17.8 °C	2.1356 - 10.678 J/cm	4.0009 - 20.004 ft-lb/in @Temperature -0.400 - 0.000 °F	Average value: 6.41 J/cm Grade Count:2
Izod Impact, Notched (ISO)	1.50 - 21000 kJ/m²	0.714 - 9990 ft-lb/in²	Average value: 9.07 kJ/m² Grade Count:148
 @Temperature -40.0 - 2.00 °C	1.10 - 62.0 kJ/m²	0.523 - 29.5 ft-lb/in² @Temperature -40.0 - 35.6 °F	Average value: 5.79 kJ/m² Grade Count:37
Izod Impact, Unnotched (ISO)	60.0 kJ/m² - NB	28.6 ft-lb/in² - NB	Average value: 80.0 kJ/m² Grade Count:8
Charpy Impact Unnotched	0.300 J/cm² - NB	1.43 ft-lb/in² - NB	Average value: 7.51 J/cm² Grade Count:155
 @Temperature -50.0 - 0.000 °C	0.480 J/cm² - NB	2.28 ft-lb/in² - NB @Temperature -58.0 - 32.0 °F	Average value: 4.31 J/cm² Grade Count:111
Charpy Impact, Notched	0.140 J/cm² - NB	0.666 ft-lb/in² - NB	Average value: 1.27 J/cm² Grade Count:259
 @Temperature -40.0 - 10.0 °C	0.100 J/cm² - NB	0.476 ft-lb/in² - NB @Temperature -40.0 - 50.0 °F	Average value: 0.524 J/cm² Grade Count:147
Gardner Impact	0.904 - 67.8 J	0.667 - 50.0 ft-lb	Average value: 18.2 J Grade Count:71
 @Temperature -30.0 - 0.000 °C	1.99983 - 63.2716 J	1.47500 - 46.6667 ft-lb @Temperature -22.0 - 32.0 °F	Average value: 21.6 J Grade Count:27
 @Temperature -30.0 - -28.9 °C	15.0225 - 32.9916 J	11.0800 - 24.3334 ft-lb @Temperature -22.0 - -20.0 °F	Average value: 21.6 J Grade Count:10

	15.0225 - 32.9916 J @Thickness 3.17 - 3.20 mm	11.0800 - 24.3334 ft-lb @Thickness 0.125 - 0.126 in	Average value: 21.6 J Grade Count:10
Falling Dart Impact	4.98 - 28.0 J	3.67 - 20.7 ft-lb	Average value: 10.8 J Grade Count:5
Instrumented Impact Total Energy	5.00 - 40.0 J	3.69 - 29.5 ft-lb	Average value: 24.9 J Grade Count:6
Tensile Creep Modulus, 1 hour	550 - 700 MPa	79800 - 102000 psi	Average value: 657 MPa Grade Count:7
Tensile Creep Modulus, 1000 hours	220 - 440 MPa	31900 - 63800 psi	Average value: 349 MPa Grade Count:7
Tear Strength	20.2 - 226 kN/m	115 - 1290 pli	Average value: 102 kN/m Grade Count:7
Compression Set	16.0 - 65.0 %	16.0 - 65.0 %	Average value: 33.2 % Grade Count:4
Film Tensile Strength at Break, MD	3.10 - 152 MPa	450 - 22000 psi	Average value: 70.1 MPa Grade Count:3
Tangent Modulus	1290 - 3260 MPa	187000 - 472000 psi	Average value: 1920 MPa Grade Count:4

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1000 - 1.00e+18 ohm-cm	1000 - 1.00e+18 ohm-cm	Average value: 1.21e+17 ohm-cm Grade Count:107
Surface Resistance	10.0 - 1.00e+15 ohm	10.0 - 1.00e+15 ohm	Average value: 9.11e+13 ohm Grade Count:96
Static Decay	0.0100 - 2.00 sec	0.0100 - 2.00 sec	Average value: 0.646 sec Grade Count:8
Dielectric Constant	2.20 - 2.38	2.20 - 2.38	Average value: 2.30 Grade Count:63
Dielectric Strength	19.7 - 140 kV/mm	500 - 3560 kV/in	Average value: 131 kV/mm Grade Count:63
Dissipation Factor	0.0000700 - 0.00300	0.0000700 - 0.00300	Average value: 0.000410 Grade Count:61
Arc Resistance	88.0 - 136 sec	88.0 - 136 sec	Average value: 104 sec Grade Count:3
Comparative Tracking Index	550 - 600 V	550 - 600 V	Average value: 598 V Grade Count:77
Tribocharge	10.0 V	10.0 V	Average value: 10.0 V Grade Count:3

Thermal Properties	Metric	English	Comments
CTE, linear	18.0 - 185 µm/m-°C	10.0 - 103 µin/in-°F	Average value: 114 µm/m-°C Grade Count:109
Melting Point	61.0 - 220 °C	142 - 428 °F	Average value: 160 °C Grade Count:266
Crystallization Temperature	110 - 115 °C	230 - 239 °F	Average value: 111 °C Grade Count:5
Maximum Service Temperature, Air	65.0 - 125 °C	149 - 257 °F	Average value: 79.4 °C Grade Count:24
Deflection Temperature at 0.46 MPa (66 psi)	37.8 - 160 °C	100 - 320 °F	Average value: 100 °C Grade Count:983
Deflection Temperature at 1.8 MPa (264	37.0 - 172 °C	98.6 - 341 °F	Average value: 62.9 °C Grade Count:291

psi)			
Vicat Softening Point	35.0 - 160 °C	95.0 - 320 °F	Average value: 122 °C Grade Count:377
Heat Distortion Temperature	85.0 - 115 °C	185 - 239 °F	Average value: 96.1 °C Grade Count:9
Minimum Service Temperature, Air	-30.0 °C	-22.0 °F	Average value: -30.0 °C Grade Count:5
Brittleness Temperature	-20.0 °C	-4.00 °F	Average value: -20.0 °C Grade Count:16
UL RTI, Electrical	65.0 - 221 °C	149 - 430 °F	Average value: 147 °C Grade Count:5
UL RTI, Mechanical with Impact	65.0 - 221 °C	149 - 430 °F	Average value: 147 °C Grade Count:5
UL RTI, Mechanical without Impact	65.0 - 221 °C	149 - 430 °F	Average value: 147 °C Grade Count:5
Flammability, UL94	HB - V-0	HB - V-0	Grade Count:111
Oxygen Index	25.0 - 27.0 %	25.0 - 27.0 %	Average value: 26.0 % Grade Count:4
Glow Wire Test	850 - 960 °C	1560 - 1760 °F	Average value: 905 °C Grade Count:4
DSC Induction Temperature	142 - 165 °C	288 - 329 °F	Average value: 159 °C Grade Count:8

Optical Properties	Metric	English	Comments
Haze	0.200 - 91.0 %	0.200 - 91.0 %	Average value: 29.5 % Grade Count:106
Gloss	30.0 - 160 %	30.0 - 160 %	Average value: 76.0 % Grade Count:39
Yellow Index	-6.00 - 2.44 %	-6.00 - 2.44 %	Average value: -0.702 % Grade Count:8
Transmission, Visible	0.500 - 99.0 %	0.500 - 99.0 %	Average value: 83.4 % Grade Count:25

Processing Properties	Metric	English	Comments
Processing Temperature	87.8 - 274 °C	190 - 525 °F	Average value: 209 °C Grade Count:56
Nozzle Temperature	190 - 270 °C	374 - 518 °F	Average value: 217 °C Grade Count:15
Melt Temperature	40.0 - 320 °C	104 - 608 °F	Average value: 225 °C Grade Count:135
Head Temperature	200 - 230 °C	392 - 446 °F	Average value: 213 °C Grade Count:4
Mold Temperature	4.00 - 91.0 °C	39.2 - 196 °F	Average value: 40.9 °C Grade Count:117
Drying Temperature	65.6 - 105 °C	150 - 221 °F	Average value: 84.1 °C Grade Count:34
Moisture Content	0.0200 - 1.00 %	0.0200 - 1.00 %	Average value: 0.290 % Grade Count:16
Injection Pressure	2.76 - 103 MPa	400 - 15000 psi	Average value: 59.5 MPa Grade Count:33
Cure Time	0.0250 - 0.0417 min	0.000417 - 0.000694 hour	Average value: 0.0333 min Grade Count:19

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.